

DBSHIELD

Project Summary:

A Java backup and recovery utility (JavaFX desktop app + optional CLI) that supports MySQL, PostgreSQL, MongoDB, and SQLite. Features include AES-256 encrypted full/incremental/differential backups, role-based access control (RBAC), scheduling, multi-destination storage (local & cloud), tamper-proof audit logs, and integrity checks.

UI Design:

1. Login Screen

- Username, Password, "Remember me", Login button
- 2FA screen popup after password: enter OTP

2. Dashboard

- Top: active user, role, quick actions (New Backup, Restore, View Logs)
- Left: sidebar (Jobs, Backups, Destinations, Users, Logs, Settings)
- Center: job list with status (Last run, Next schedule, Size)

3. New Backup Job

- Form: Job name, DB type, Connection info, Tables to include/exclude, Mask sensitive fields (checkboxes), Destination (local/cloud), Encryption (yes/no), Schedule options
- Buttons: Test Connection, Save, Run Now

4. Restore Screen

- Select backup (from list or by ID), restore options, 2FA prompt for restores

5. Logs / Audit

- Filterable table: time, user, operation, status, SHA-256 hash, confirm signature

6. Settings

- Keystore management, Cloud credentials management, Retention policy UI

-- Database Connections

```
CREATE TABLE db_connections(  
  
    conn_id INTEGER PRIMARY KEY AUTOINCREMENT,  
  
    name VARCHAR(100) NOT NULL,  
  
    db_type VARCHAR(20) NOT NULL,    -- mysql/postgres/mongo/sqlite  
  
    host VARCHAR(255),  
  
    port INTEGER,  
  
    database_name VARCHAR(100),  
  
    username VARCHAR(100),  
  
    encrypted_password VARBINARY(512), -- encrypted with keystore  
  
    connection_string TEXT,          -- optional full DSN/URI  
  
    is_active BOOLEAN DEFAULT TRUE,  
  
    created_by INTEGER,  
  
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,  
  
    FOREIGN KEY(created_by) REFERENCES users(user_id)  
  
);
```

-- Backup Jobs

```
CREATE TABLE backup_jobs(  
  
    job_id INTEGER PRIMARY KEY AUTOINCREMENT,  
  
    job_name VARCHAR(100) NOT NULL,  
  
    conn_id INTEGER NOT NULL,  
  
    include_tables TEXT,            -- JSON or comma list  
  
    exclude_tables TEXT,
```

```

mask_rules TEXT,          -- JSON describing masking
dest_type VARCHAR(20),    -- local/aws/gcp/azure
dest_config TEXT,         -- JSON for bucket/path
schedule_cron VARCHAR(100), -- cron expression
retention_days INTEGER DEFAULT 30,
compression_alg VARCHAR(30) DEFAULT 'gzip',
encryption_required BOOLEAN DEFAULT TRUE,
status VARCHAR(20) DEFAULT 'ACTIVE', -- ACTIVE/PAUSED/DELETED
created_by INTEGER,
created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY(conn_id) REFERENCES db_connections(conn_id),
FOREIGN KEY(created_by) REFERENCES users(user_id)
);

```

-- Backup Records

```

CREATE TABLE backup_records(
    record_id INTEGER PRIMARY KEY AUTOINCREMENT,
    job_id INTEGER NOT NULL,
    timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
    file_path VARCHAR(500),
    file_size BIGINT,
    checksum VARCHAR(128),    -- SHA-256
    encryption_alg VARCHAR(50), -- AES-256-GCM
    compression_alg VARCHAR(30),

```

```

duration_seconds INTEGER,

status VARCHAR(20),          -- SUCCESS/FAILED

restore_status VARCHAR(20),   -- SUCCESS/FAILED/NOT_ATTEMPTED

log TEXT,

created_by INTEGER,

FOREIGN KEY(job_id) REFERENCES backup_jobs(job_id),

FOREIGN KEY(created_by) REFERENCES users(user_id)

);

```

-- Audit Logs

```

CREATE TABLE audit_logs(

audit_id INTEGER PRIMARY KEY AUTOINCREMENT,

event_time DATETIME DEFAULT CURRENT_TIMESTAMP,

user_id INTEGER,

job_id INTEGER,

event_type VARCHAR(50),      -- BACKUP_START, RESTORE_END etc.

resource VARCHAR(200),

detail TEXT,

ip_address VARCHAR(100),

signature VARBINARY(512),    -- digital signature

FOREIGN KEY(user_id) REFERENCES users(user_id),

FOREIGN KEY(job_id) REFERENCES backup_jobs(job_id)

);

```

-- Keystore Metadata

```
CREATE TABLE keystore_meta(  
  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  
  key_alias VARCHAR(100),  
  
  algorithm VARCHAR(50) DEFAULT 'AES-256',  
  
  is_active BOOLEAN DEFAULT TRUE,  
  
  rotation_date DATETIME,  
  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
  
);
```

TECH STACK:

- **Language / Runtime:** Java 17 (LTS)
- **Build Tool:** Maven (fat JAR with Shade, installers via jpackage)
- **GUI:** JavaFX (thin controllers, no business logic)
- **CLI:** Picocli (optional, for automation/DevOps use)
- **Database Drivers:** JDBC (MySQL/PostgreSQL/SQLite), MongoDB Java Driver
- **Metadata Store:** SQLite (single-user) or MySQL (multi-user/team)
- **Scheduler:** Quartz Scheduler
- **Crypto:** BouncyCastle / JCE (AES-256-GCM, SHA-256)
- **Cloud SDKs:** AWS SDK v2 (S3), Google Cloud Storage SDK, Azure Blob SDK
- **Logging:** SLF4J + Logback
- **Testing:** JUnit 5 + Testcontainers (real DB integration tests)
- **Packaging & Deployment:** Dockerfile, GitHub Actions CI/CD
- **Version Control:** Git + GitHub

ERD:

